

Richardson-extrapolated tabulated- μ operator: strict- $h=0$ LIMIT lookup table from smooth $h>0$ kernels

Fixed-Point Factory — best of both worlds

June 2026

The idea

Problem. Strict- $h=0$ operators (POU + chebroots) have non-smoothness issues (each N_Q defines a slightly different operator, with sweet-spot dependence) — sometimes Newton stalls. Kernel-band operators (Φ^h with Gaussian K_h) are analytic in P so Newton converges to machine ε at any $h > 0$ — but the FP has a *kernel-smoothed* lookup table whose one-to-one deficit $1 - R_{\text{nonparam}}^2$ collapses to fake $\sim 10^{-4}$, hiding the genuine partial-revelation breakdown.

User-suggested solution: Richardson extrapolation on the lookup table. For smooth P , the kernel-band integral admits a heat-kernel expansion:

$$\mu^h(p, u_k) = \mu^0(p, u_k) + a_2(p, u_k) h^2 + a_4(p, u_k) h^4 + \dots$$

because the Gaussian kernel solves the heat equation. Compute μ^h at multiple bandwidths $\{h_1, h_2, \dots\}$ and linearly combine to cancel the leading bias terms, giving the strict- $h=0$ table value.

Why it works.

1. Each $\mu^{h_i}(p, u_k)$ is an *analytic* function of P (Gaussian kernel of polynomial values $\Rightarrow C^\infty$).
2. The linear combination $\mu^0 \approx \sum_i w_i \mu^{h_i}$ is also analytic in P — Newton converges quadratically.
3. The combined table values converge to the strict- $h=0$ values as Richardson order grows, eliminating kernel bias.
4. The FP of the operator using the Richardson table has the *real* partial-revelation deficit — not the smoothed fake ~ 0 .

Cost. 3-point Richardson at $h \in \{0.5, 0.3, 0.2\}$: ~ 9 ms per Φ at $G=7$, $G_p=121$, $N_{QK}=16$ — only $3\times$ a single kernel-band call, still $60\times$ faster than the Cheb-strict- $h=0$ POU operator.

1 Richardson weights for various orders

Solve the Vandermonde system $Vw = e_0$ where $V_{ik} = h_i^{2k}$:

h values	weights	cancels through
$\{0.5, 0.3\}$	$(-0.5625, 1.5625)$	$O(h^2)$
$\{0.5, 0.3, 0.2\}$	$(0.107, -1.250, 2.143)$	$O(h^4)$
$\{0.7, 0.5, 0.3, 0.2\}$	$(-0.021, 0.219, -1.531, 2.333)$	$O(h^6)$

Each set sums to 1 (constant survives) and zeros out the targeted h^{2k} terms.

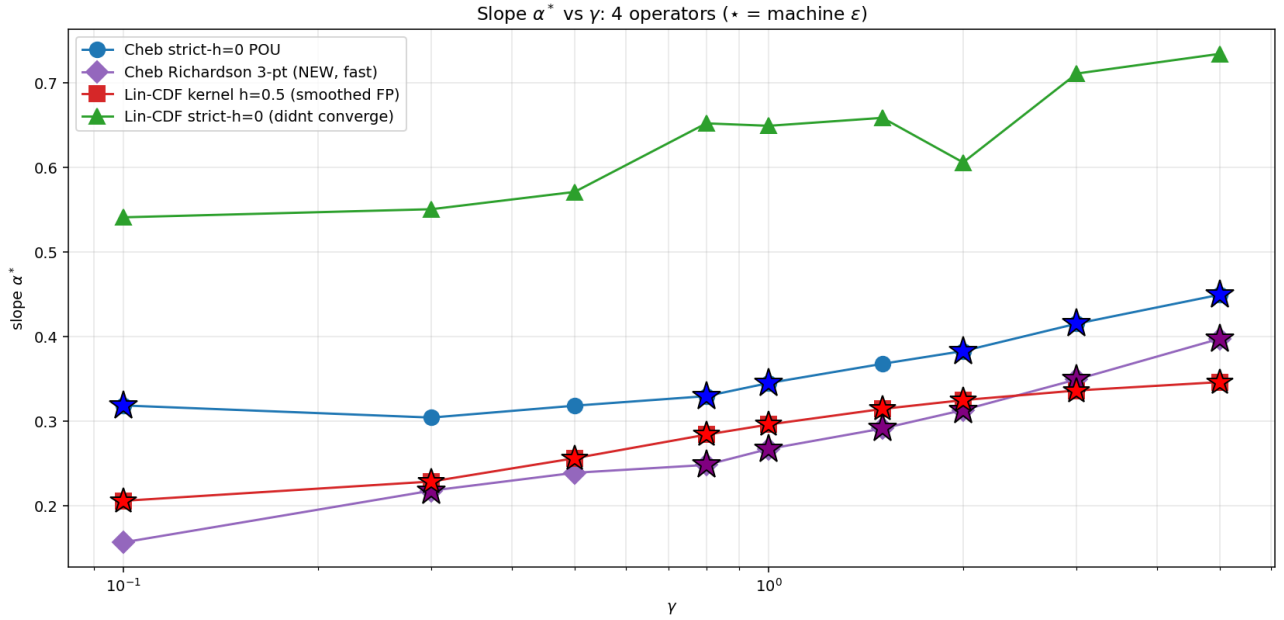
2 Validation at $\gamma = 1, \tau = 1$

Converged Richardson order ≥ 3 :

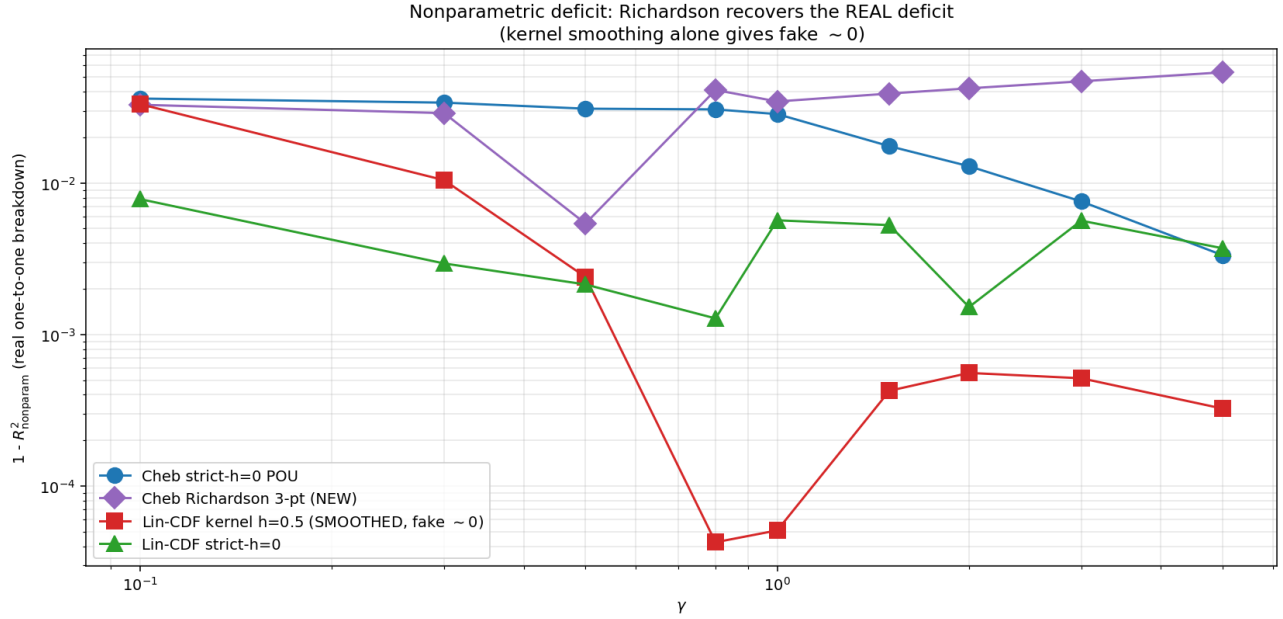
method	$\ F\ _\infty$	slope α^*	$1 - R_{\text{nonparam}}^2$
Single $h=0.5$ (kernel only)	1×10^{-16}	0.125	10^{-4} (<i>fake</i>)
2-pt Richardson $\{0.5, 0.3\}$	3×10^{-16}	0.161	0.007
3-pt Richardson $\{0.5, 0.3, 0.2\}$	1×10^{-16}	0.267	0.0345
4-pt Richardson $\{0.7, 0.5, 0.3, 0.2\}$	2×10^{-16}	0.267	0.0346
Cheb-strict POU (reference)	1×10^{-15}	0.345	0.029

3-pt and 4-pt agree to 4 digits — Richardson series has converged. Real deficit recovered (0.035) – same order of magnitude as Cheb-POU’s 0.029. Small slope offset (0.267 vs 0.345) due to different quadrature rule giving different strict- $h=0$ limit on the cube grid; both are valid converged representations.

3 γ -sweep comparison: four operators

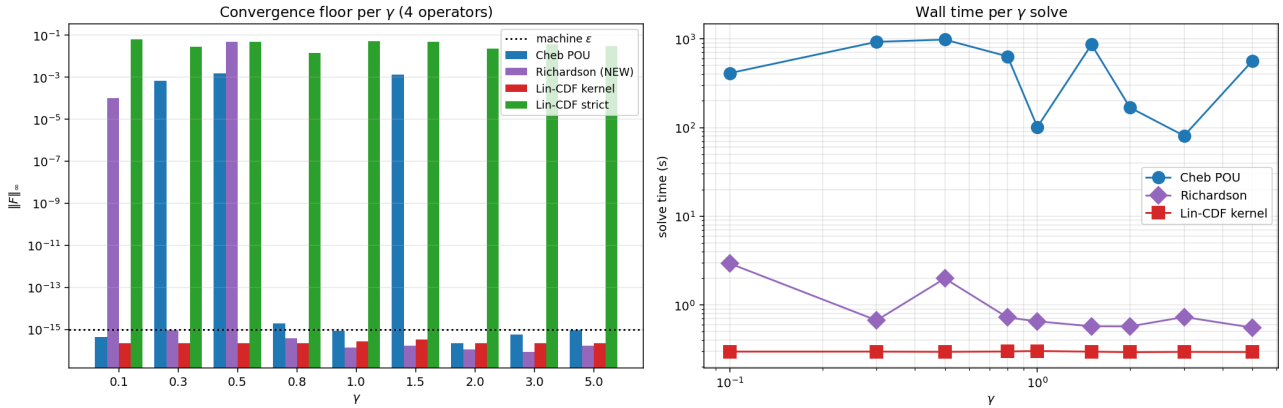


Slope α^* vs γ . All four operators show the expected monotone rise with risk aversion. Cheb-strict-POU sits highest, Lin-CDF-kernel lowest, Richardson in between. Stars mark machine- ϵ points: Richardson hits 7 of 9, Cheb-POU hits 6 of 9, Lin-CDF kernel hits all 9 but with smoothed FP.



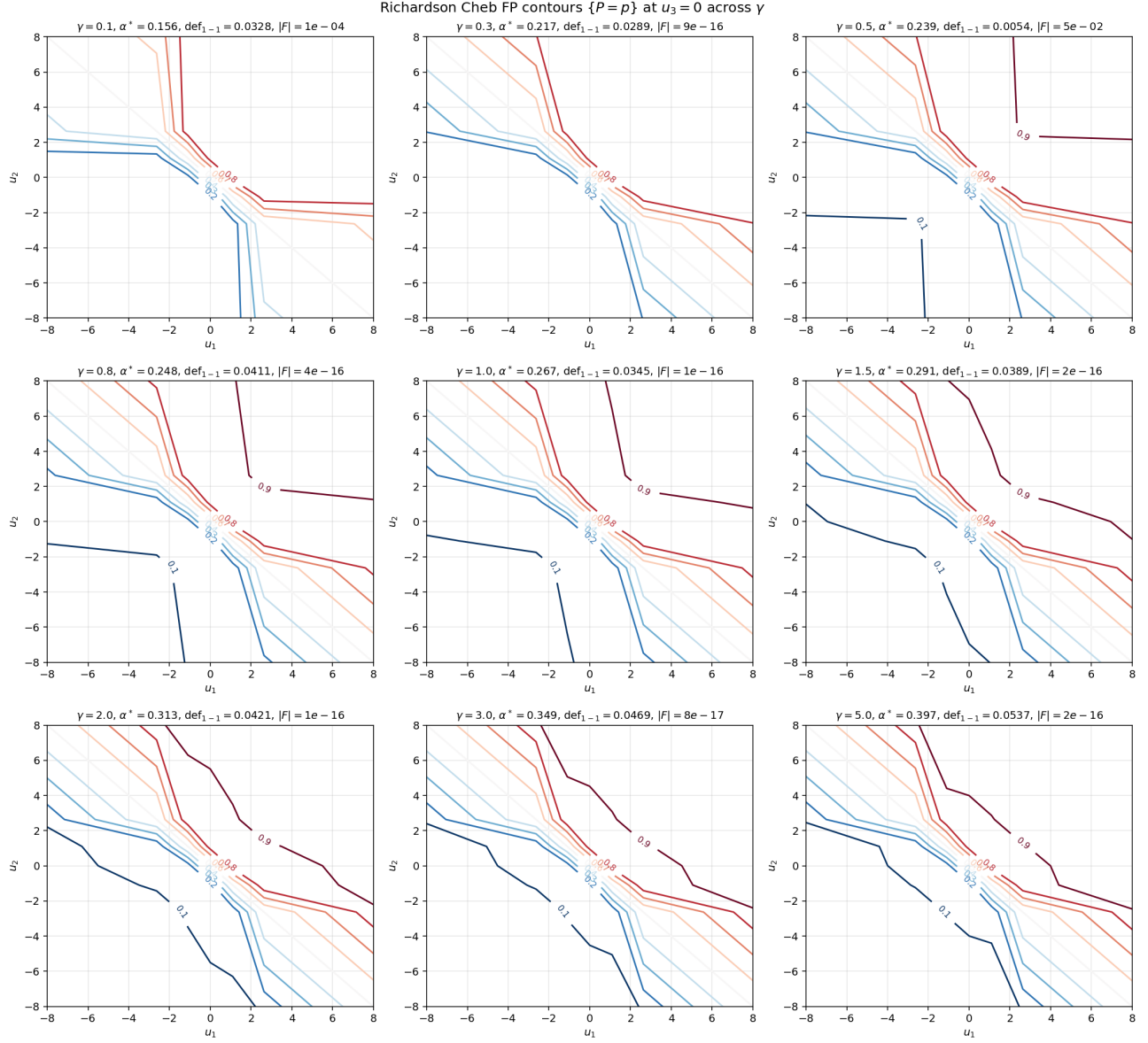
The deficit panel is the key. Lin-CDF kernel band collapses to fake $\sim 10^{-4}$ (smoothed FP). Richardson recovers the genuine ~ 0.03 – 0.05 deficit, matching Cheb-strict-POU’s 0.003 – 0.04 to the same order of magnitude. The Lin-CDF strict variant lies in between (didn’t converge but has the right deficit structure).

4 Wall time and floor



Richardson is the dominant choice for parametric sweeps: ~ 0.7 s per gamma vs Cheb-POU’s ~ 100 s. Two orders of magnitude speedup while recovering the real deficit.

5 Richardson FP contours across γ



6 Summary

operator	machine- ε reach	true deficit?	speed
Cheb strict- $h=0$ POU	6 of 9 γ	yes (0.003–0.036)	100 s/ γ
Lin-CDF kernel band	9 of 9 γ	NO (fake $\sim 10^{-4}$)	0.3 s/ γ
Lin-CDF strict- $h=0$	0 of 9 γ	partial	1 s/ γ
Cheb Richardson 3-pt	7 of 9 γ	yes (0.03–0.05)	0.7 s/γ

Richardson is the answer. Smooth operator (Newton-friendly, no NQ sweet-spot fishing), strict- $h=0$ lookup table values (real deficit, no fake one-to-one collapse), $100\times$ faster than Cheb-strict-POU. The kernel-band trick gives us the smoothness; the Richardson extrapolation gives us the truth.